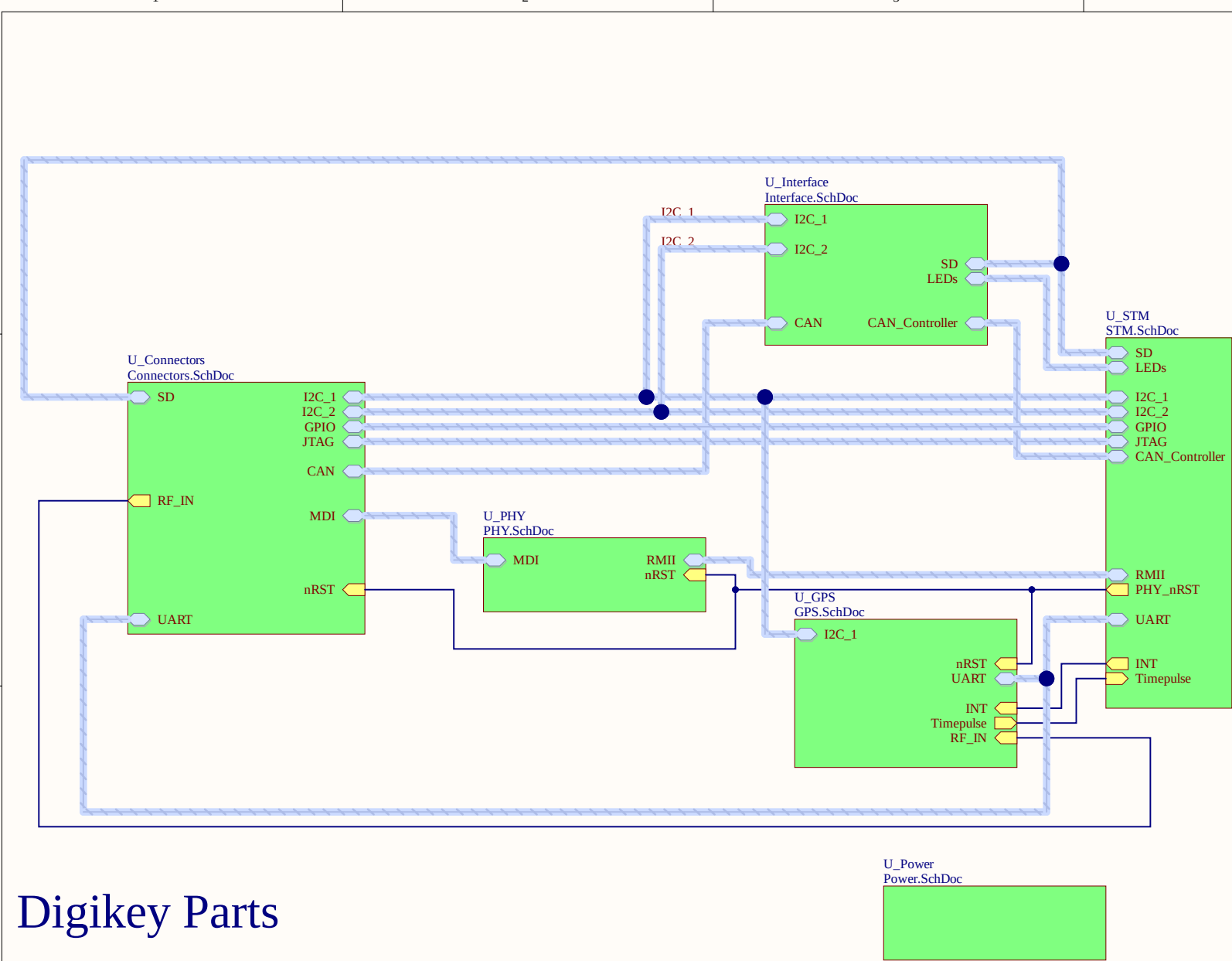




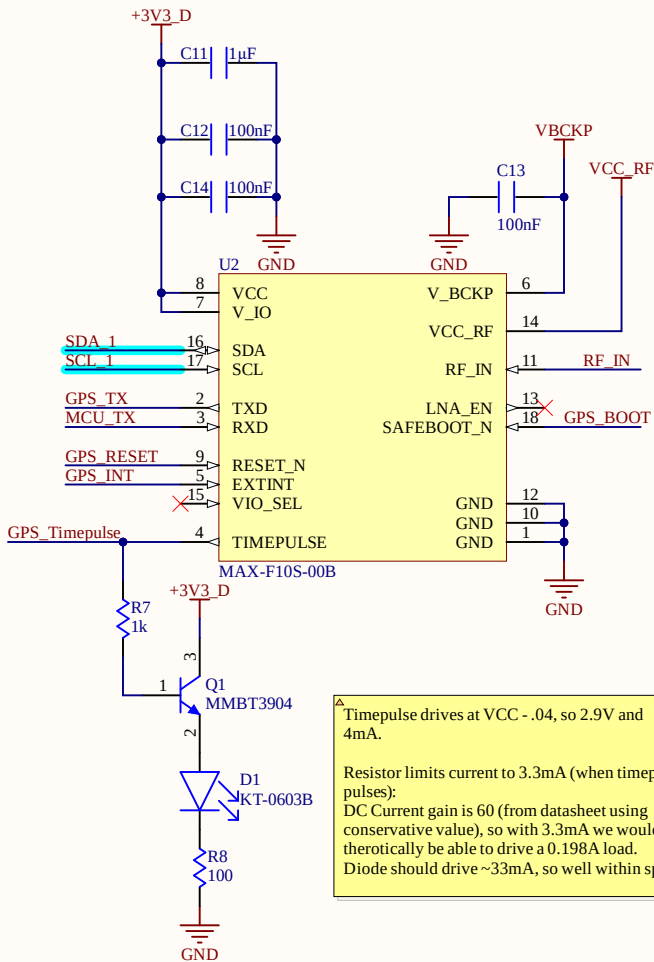
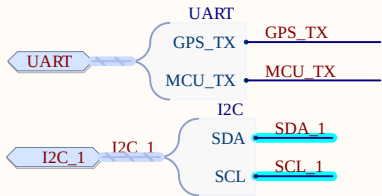
Digikey Parts

Desig.	Part Name	Desig.	Part Name
J1	3 Pin Header	U4	CAN Transciever
J2	34 Pin Ampseal		
J4	2 Pin Header		
J5	2 Pin Header		
J6	2 Pin Header		
J8	2 Pin Header		
J9	2 Pin Header		
J11	8 Pin Header		
BT1	1056TR		

Rev.	Events, Revisions, Details	Date
V1.0	Design completed to requirements.	08/05/2027

PROJECT	Digital Board			TITLE	Top Level		
Last Edit:	8/19/2026			CAR	MF14	SIZE	B
Date:	8/29/2026			REV	1		
Part Num:	ELC-1-DIG-1			Mines Formula SAE Electrical			
Drawn By:	Grant Uher			SHEET NUMBER 1 OF 8			
Approved By:							

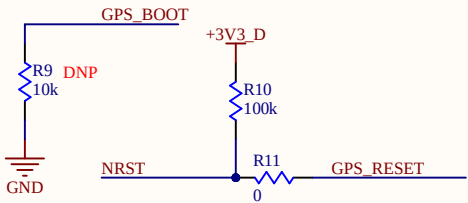




Timepulse drives at VCC - .04, so 2.9V and 4mA.

Resistor limits current to 3.3mA (when timepulse pulses):
DC Current gain is 60 (from datasheet using conservative value), so with 3.3mA we would therotically be able to drive a 0.198A load.
Diode should drive ~33mA, so well within specs.

15	VIO_SEL	-	I	Voltage selector for V_IO supply. Connect to GND for 1.8 V supply, or leave open for 3.3 V supply.
16	ENA	2	HP	RF data 1 enable (not used)
18	SAFEBOOT_N	-	I	Safeboot mode (active low). Leave open if not used. ¹⁴



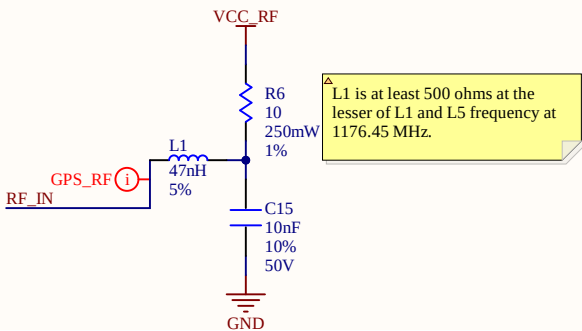
CONFIG

C.1 Standard capacitors

Table 40 presents the recommended capacitor values for MAX-F10S.

Name	Use	Type / Value
C14	RF Bias-T capacitor	10 nF, 10%, 16 V, X7R

Table 40: Standard capacitors

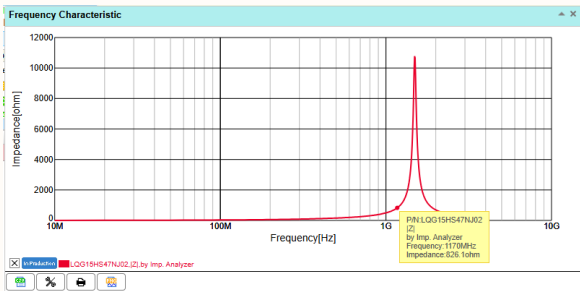


C.3 Inductors

Table 42 presents the recommended inductor values for MAX-F10S.

Name	Use	Type / Value	Recommended component
L4	RF Bias-T inductor	47 nH, 5%	Murata LQG15HS47NJ02 Johanson Technology L-07W series Any other inductor with impedance > 500 Ω at GNSS L1 and L5 frequencies and current rating above 300 mA.

Table 42: Recommended inductors



PROJECT	Digital Board	TITLE	GPS	MINES FORMULA
Last Edit:	8/19/2026	CAR	MF14	
Date:	8/29/2026	SIZE	B	
Part Num:	ELC-1-DIG-1	REV	1	
Drawn By:	Grant Uher	SHEET NUMBER	3 OF 8	

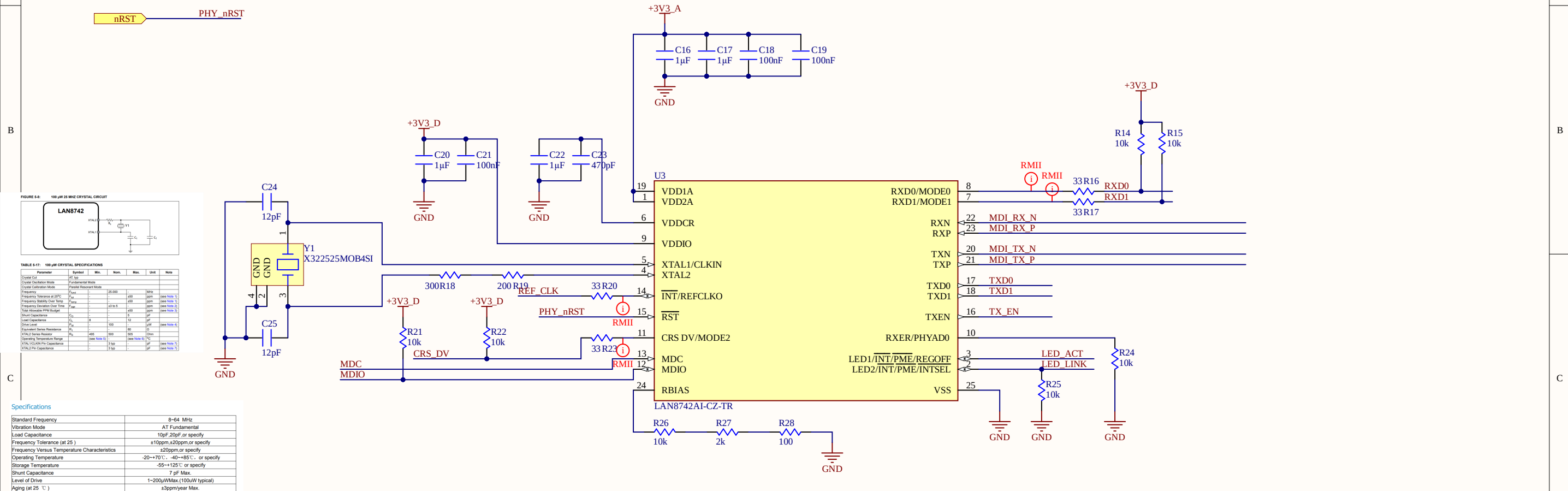
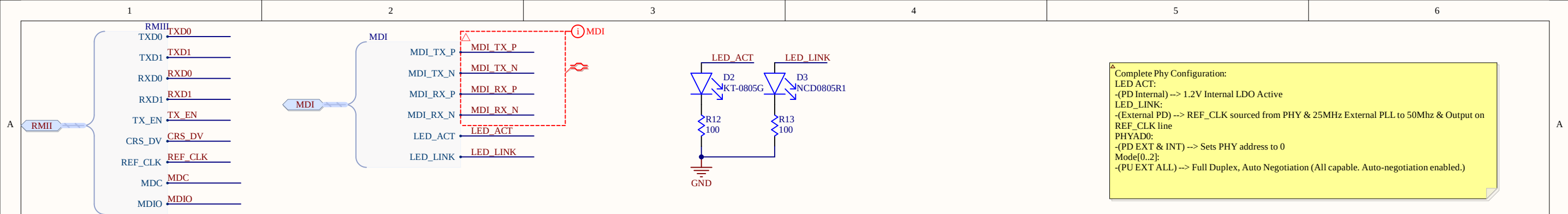


FIGURE 3-15: LED2/nINTSEL POLARITY CONFIGURATION

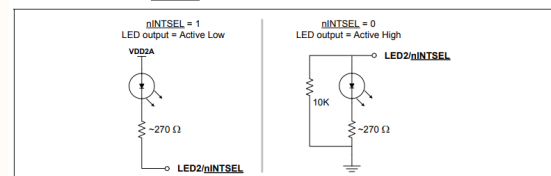
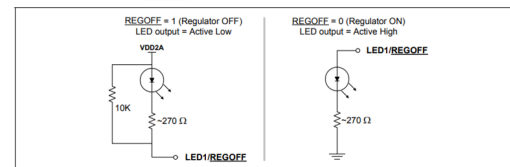
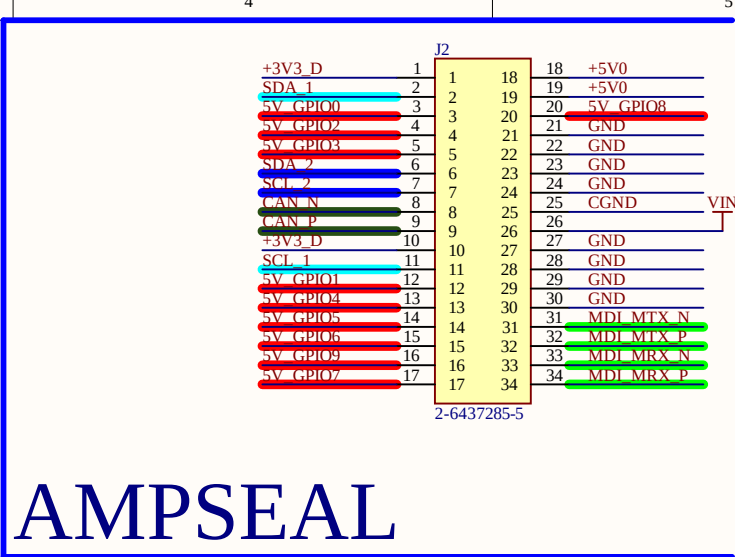
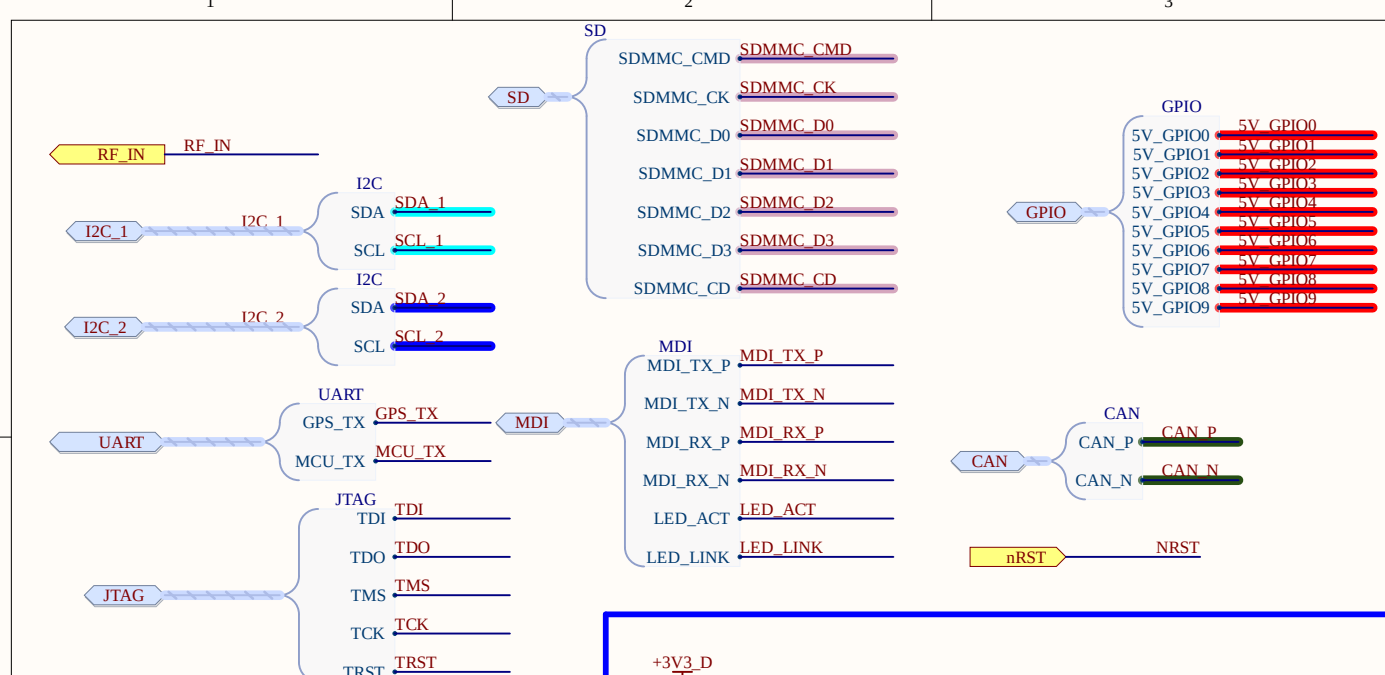


FIGURE 3-14: LED1/REGOFF POLARITY CONFIGURATION

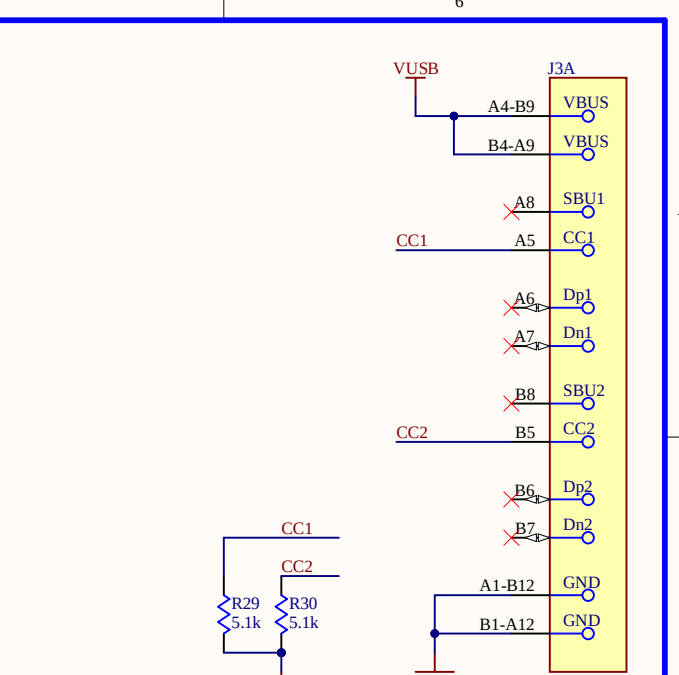
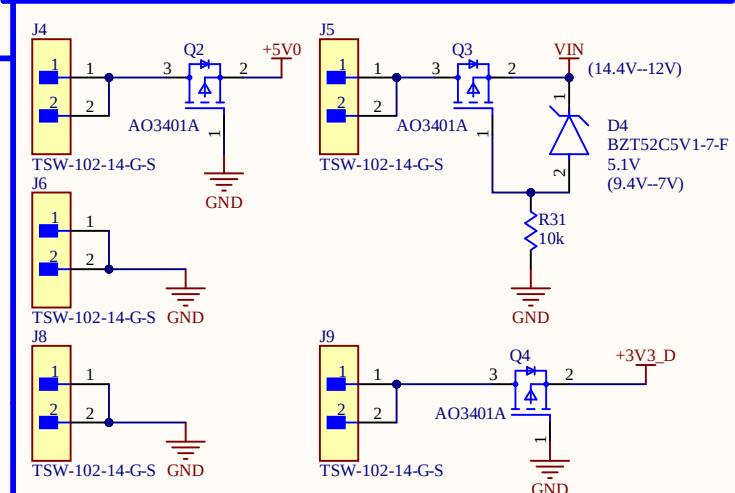


PROJECT	TITLE			
Digital Board	Ethernet PHY			
Last Edit: 8/19/2026	CAR	SIZE	REV	
Date: 8/29/2026	MF14	B	1	
Part Num: ELC-1-DIG-1	Mines Formula SAE Electrical			
Drawn By: Grant Uher	SHEET NUMBER 4 OF 8			
Approved By:				

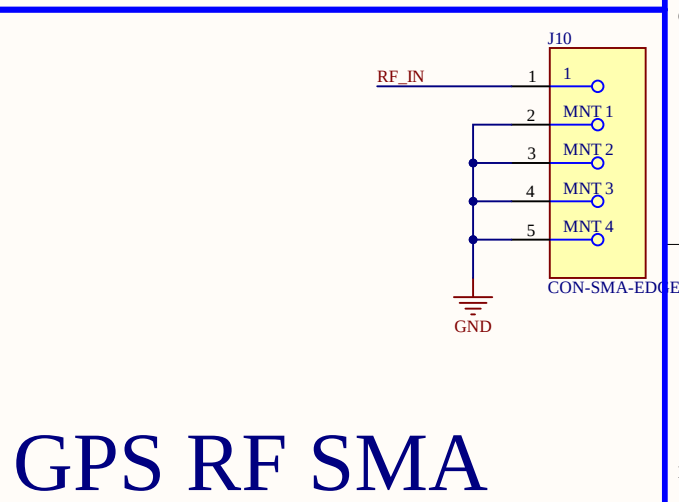




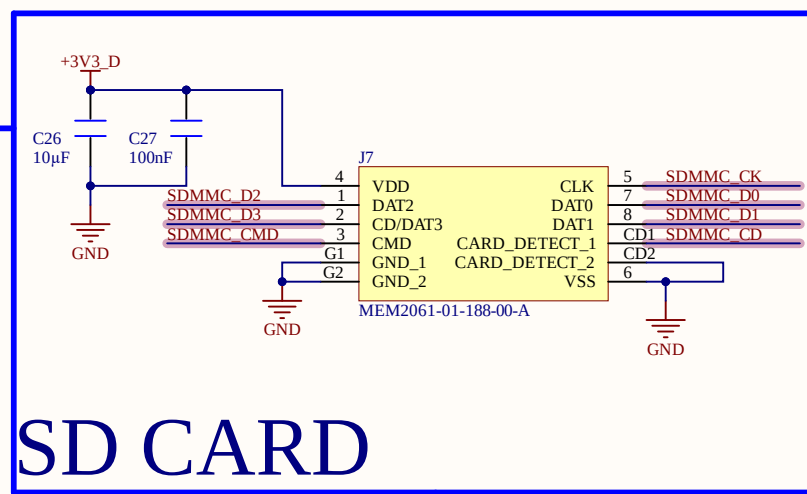
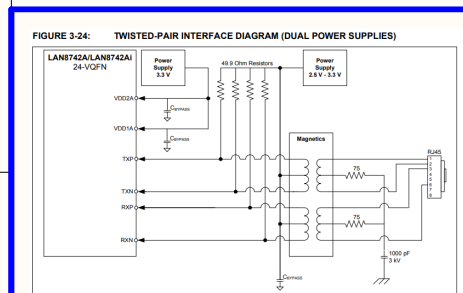
AMPSEAL



USB C Power

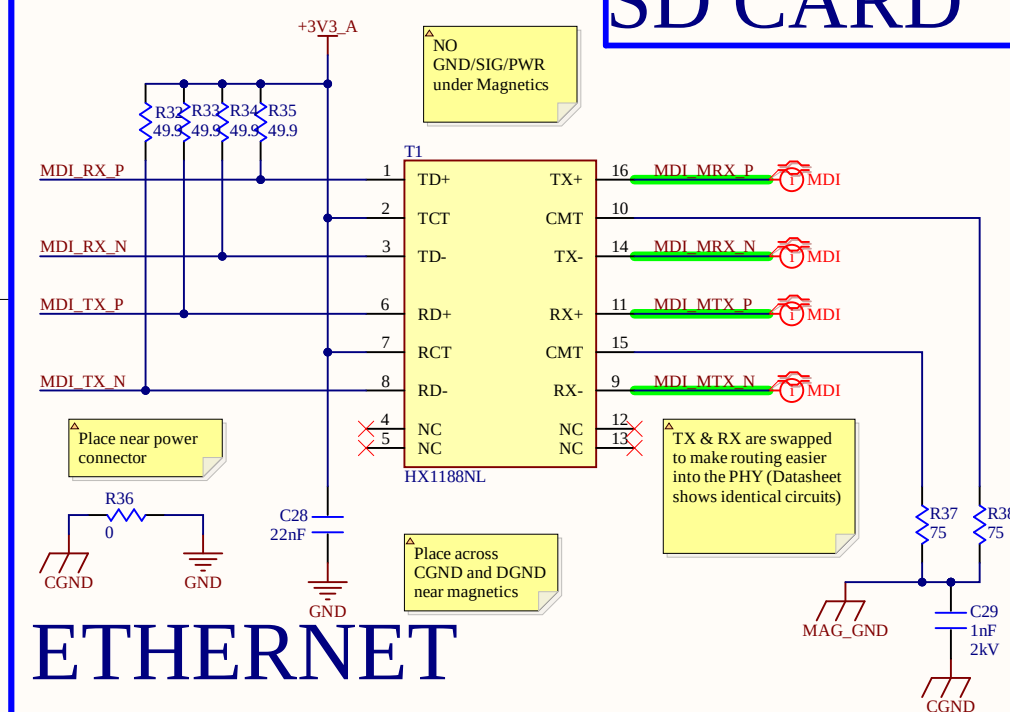


GPS RF SMA



SD CARD

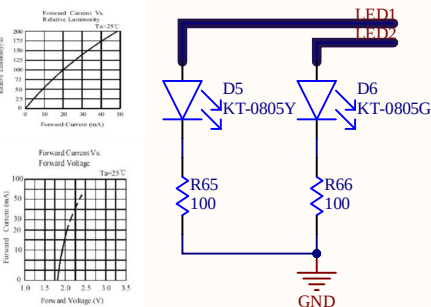
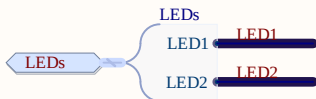
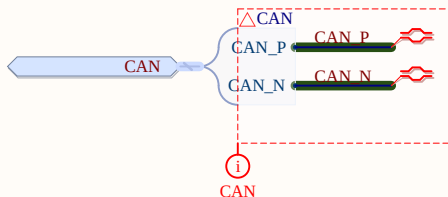
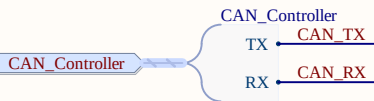
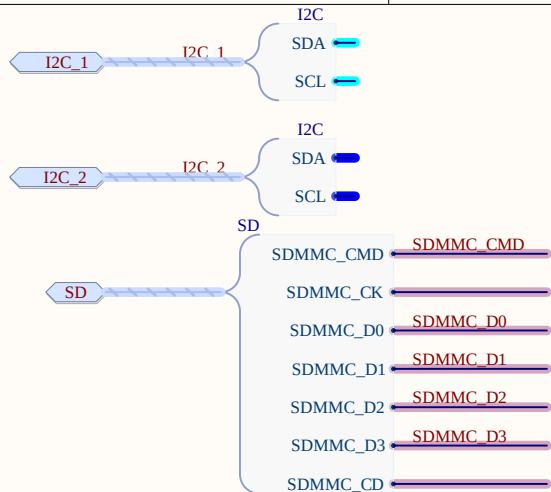
TEST POINTS



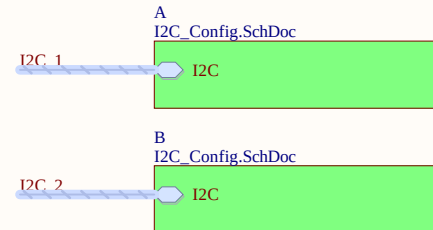
PROGRAM

PROJECT	Digital Board	TITLE	Connectors
Last Edit:	8/19/2026	CAR	MF14
Date:	8/29/2026	SIZE	B
Part Num:	ELC-1-DIG-1	REV	1
Drawn By:	Grant Uher	Mines Formula SAE Electrical	
Approved By:		SHEET NUMBER	5 OF 8

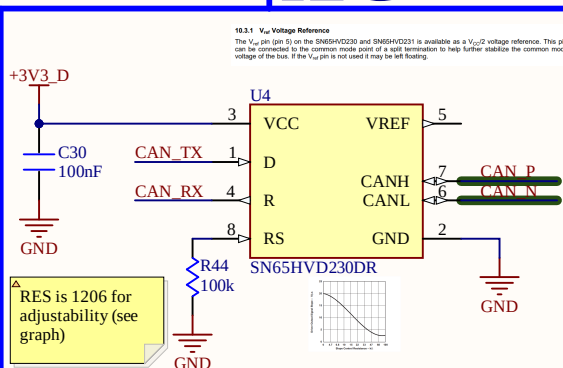
MINES FORMULA



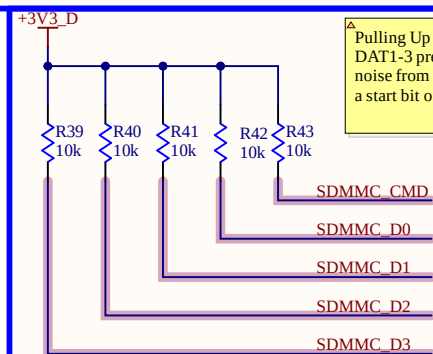
LEDs



I2C



CAN DNP

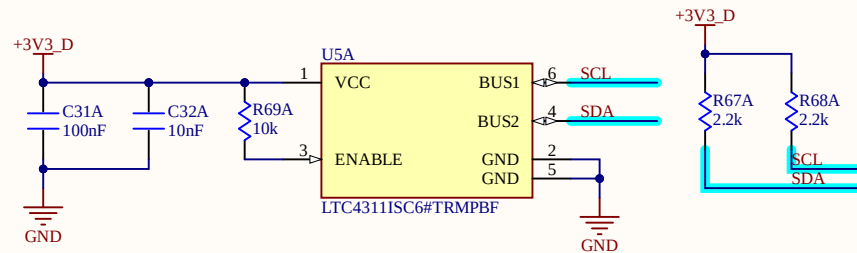
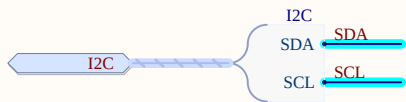


Pulling Up
DAT1-3 prevents
noise from calling
a start bit of low.

SD Pull Ups

PROJECT	Digital Board			TITLE	Interface		
Last Edit:	8/19/2026			Date:	8/29/2026		
Part Num:	ELC-1-DIG-1			CAR	MF14	SIZE	A
Drawn By:	Grant Uher			Mines Formula SAE Electrical	REV	1	
Approved By:				SHEET NUMBER	6 OF 8		

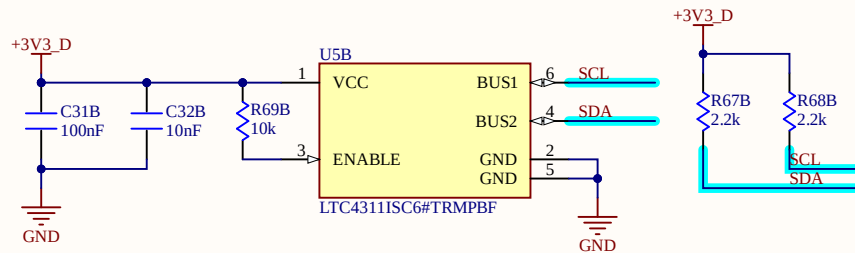
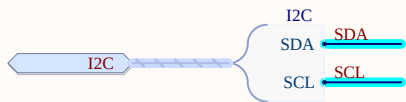




Place 10nF closer than 100nF caps

PROJECT	TITLE		
Digital Board	I2C Range Extender		
Last Edit: 8/19/2026	CAR	SIZE	REV
Date: 8/29/2026	MF14	A	1
Part Num: ELC-1-DIG-1	Mines Formula SAE Electrical		
Drawn By: Grant Uher	SHEET NUMBER 7 OF 8		
Approved By:			





Place 10nF closer than 100nF caps

PROJECT	Digital Board			TITLE	I2C Range Extender			
Last Edit:	8/19/2026			CAR	MF14	SIZE	A	
Date:	8/29/2026			REV	1			
Part Num:	ELC-1-DIG-1			Mines Formula SAE Electrical				
Drawn By:	Grant Uher			SHEET NUMBER 7 OF 8				
Approved By:								

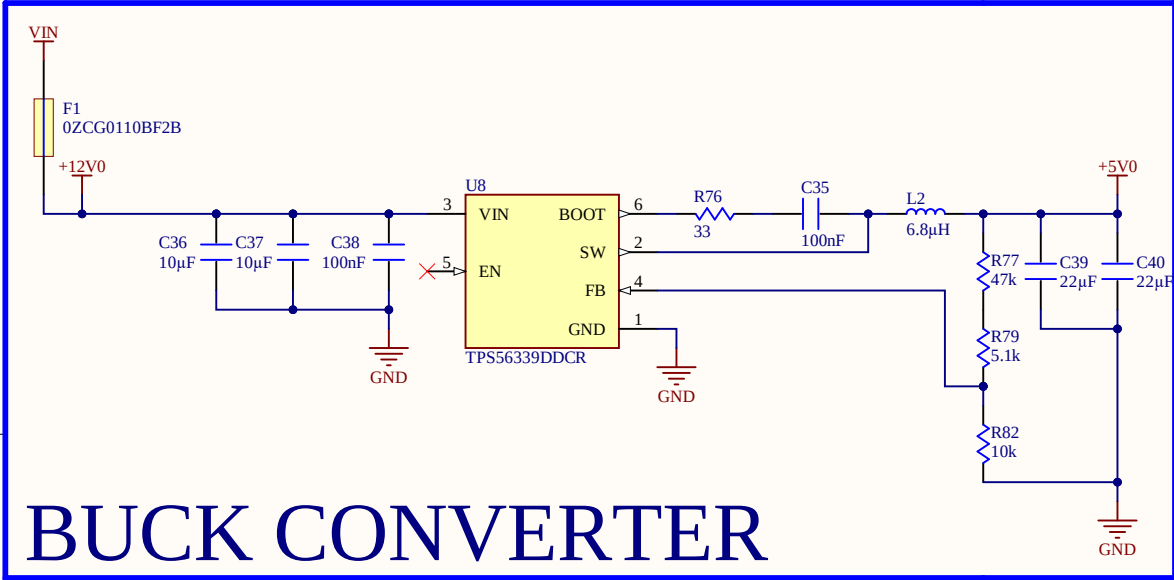


Power Budget:
12V:
LED: 12mA

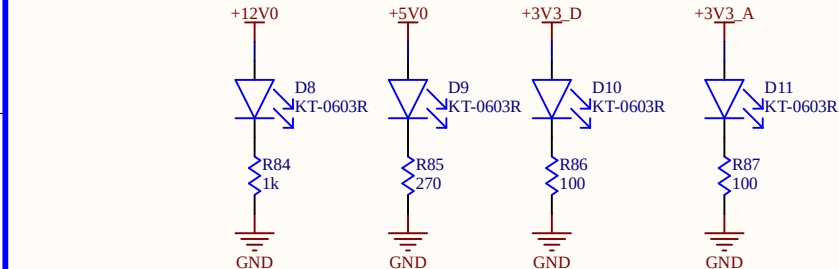
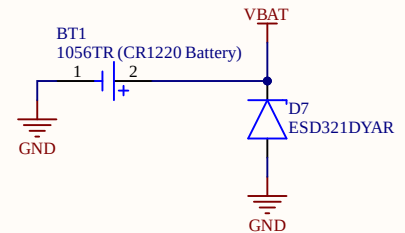
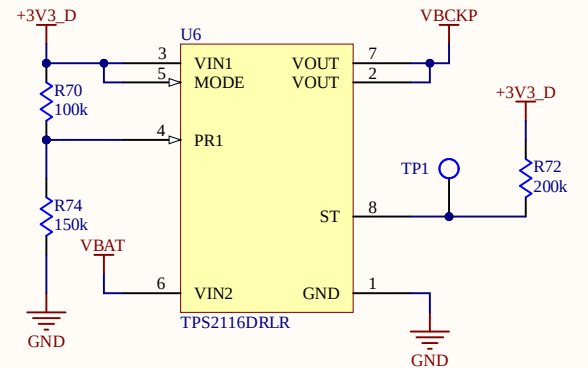
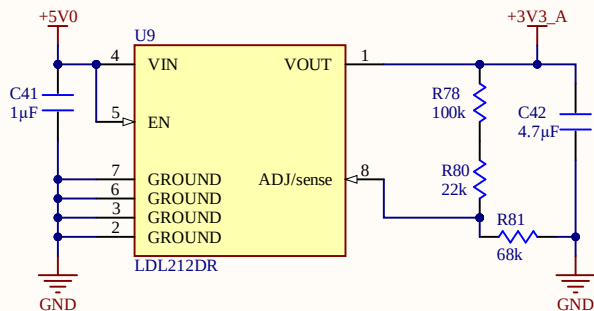
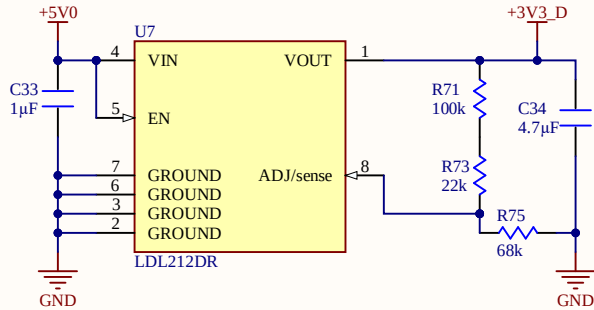
5V:
LED: 18mA

3.3V:
STM32: 120mA
NEO-M9N: 100mA
LAN8742A: 140mA
LEDs x2: 33mA

Total: around 450mA

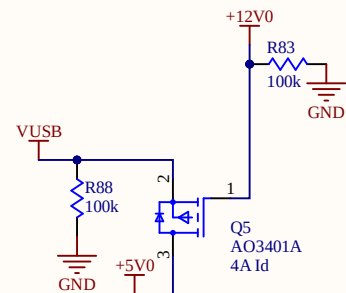


BUCK CONVERTER



LEDs

LDOs



USB Backpower Prot

GPS BACKUP

PROJECT		TITLE			
Digital Board		Power			
Last Edit: 8/19/2026		CAR	SIZE	REV	
Date: 8/29/2026		MF14	B	1	
Part Num: ELC-1-DIG-1		Mines Formula SAE Electrical			
Drawn By: Grant Uher		SHEET NUMBER 8 OF 8			
Approved By:					